Local Affine Subtraction Reveals Persistent Tangential Non-Affine Activity in Laboratory Midge Swarms

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***Abstract*—Laboratory midge swarms remain cohesive without strong global velocity alignment, making them useful for asking which components of collective motion remain after ordinary local geometry is removed. We re-analyzed nineteen three-dimensional trajectories of *Chironomus riparius* swarms and defined T1 as a local tangential non-affine activity computed from focal-neighborhood relative displacements after local affine deformation was fitted and subtracted. Under a frozen centered-detrending definition and a two-scale consistency requirement, T1 survived in 14 of 19 observations. A completed full-pipeline pseudo-event calibration found that none of 1000 null replicates reached this count (plus-one empirical $p \approx 0.001$), and local-affine conditioning quality control found well-conditioned fits. However, a detrending challenge reduced both-scale support to 11 of 19 under past-only detrending and 13 of 19 without rolling detrending, so the exact 14 of 19 count should not be read as preprocessing-invariant. Within the survivor class, support remained stable under nearby scale and temporal-lag perturbations. The most repeatable phenotype was diffuse tangential activity, whereas near-pre timing, edge/core contrast, and signed event-type structure were weaker or more heterogeneous. Compact state variables did not provide stable grouped out-of-sample improvement for first- or second-moment prediction, and true transition timestamps did not add robust near-pre excess after state-matched controls. Recent state-path history was informative in some observations but did not define a universal memory rule. These results identify T1 as a measurable local residual layer between raw individual trajectories and low-dimensional swarm state. The contribution is therefore diagnostic rather than mechanistic: T1 is common under the tested local affine reduction, but not universal or fully explained by the reductions examined here.**

***Index Terms*—midge swarms, non-affine motion, local affine subtraction, collective motion, compact state variables, observation heterogeneity**

I. Introduction

Collective motion is often studied through systems that display visible global order, such as polarized schools, flocks, or moving groups. Laboratory midge swarms provide a different test case. The insects move in a disordered three-dimensional cloud and do not exhibit a strong global orientational order, yet the swarm remains spatially cohesive and shows reproducible internal structure. This combination raises a basic question: at what level should the organization of such a weakly ordered group be described?

The question is not trivial because position-dependent velocity differences are expected even in the absence of a new behavioral mechanism. Translation of the swarm center, local rotation, expansion, compression, shear, and density gradients can all make insects at different positions move differently. Thus, observing that local velocities differ across the swarm is not by itself a strong result. A more informative test is to ask whether any stable component of motion remains after ordinary local kinematic explanations have been given a clear baseline.

A. Existing Descriptions of Midge Swarms

Previous work has shown that laboratory midge swarms are not uniform random particle clouds. Time-resolved three-dimensional trajectory data provide the basis for measuring individual motion, local neighborhoods, and global swarm geometry [8]. Individuals also sample the swarm volume non-uniformly, so spatial position and local density must be treated as potential contributors to any local motion pattern [1].

A second line of work has described swarms using macroscopic and statistical-physics variables. Swarm size, radius, density, compactness, and layer structure provide compact summaries of the full trajectory data, and equation-of-state and thermodynamic descriptions show that weakly ordered swarms can nevertheless possess reproducible collective states [7], [9]. These studies make low-dimensional state variables natural explanatory candidates for local motion, but they do not by themselves determine whether a local residual can be accounted for by those variables.

A third line of work treats midge motion in stochastic dynamical terms. Restoring-force descriptions, Langevin-type models, and intrinsic stochasticity have been used to connect microscopic fluctuations with macroscopic swarm organization [3]–[5]. This perspective is important for the present study because a residual signal need not imply a deterministic rule. It may reflect incomplete geometric subtraction, a compact stochastic state description, or local organization that is not captured by the tested variables.

Finally, perturbation and correlation analyses show that midge swarms contain measurable spatiotemporal structure. Environmental perturbations can induce correlated responses, and spatial correlations have been reported in laboratory insect swarms [2], [6]. Such correlations motivate a local analysis, but they do not distinguish local coordination from ordinary geometric deformation, sampling structure, or density-dependent motion. This leaves open the specific residual question addressed here.

B. Non-Affine Residuals as a Methodological Reference Point

A useful methodological precedent comes from non-affine analysis in complex many-body systems. In amorphous solids and colloidal glasses, local affine motion is often fitted first, and the remaining non-affine displacement is then used to characterize local rearrangement [10], [11]. Related decompositions have also been adapted to biological collective motion, including epithelial sheet migration and cell assemblies, where large-scale collective motion must be separated from local rearrangements [12], [13].

The role of this literature in the present paper is methodological rather than ontological. We do not treat a midge swarm as a glass, granular material, or elastic continuum. Instead, we borrow a more general diagnostic logic: first allow a local neighborhood to explain as much relative motion as possible through a smooth affine deformation field, and then analyze only the part that remains. This choice makes the baseline conservative. Local rotation, extension, compression, and shear are not immediately interpreted as new collective organization; they are first absorbed by the affine fit.

C. From Candidate Explanations to T1

The present analysis was motivated by a sequence of candidate explanations. First, we asked whether the transition-linked tangential signal could be absorbed by ordinary geometry, including global motion and local affine deformation. If local affine geometry removed the signal, no additional local residual object would be needed. Second, when a residual remained, we asked whether it was stable to nearby choices of local scale and temporal lag. Third, we examined whether the residual had a repeated spatial or temporal form, such as an edge-localized contrast, a sharp event precursor, or a signed transition pattern. Finally, we tested whether a small set of state variables, true event timing, or recent state-path history was sufficient to account for it.

This paper therefore does not start from a proposed complete mechanism. It starts from a residual object that survives several natural explanations. For a focal individual i at time t, we describe each retained neighbor $j \in N_i(t)$ by the focal-centered relative vector $\mathbf{r}_{ij}(t) = \mathbf{x}_j(t) - \mathbf{x}_i(t)$. Over a finite lag ℓ, the neighborhood deformation is summarized by $\Delta\mathbf{r}_{ij}(t,\ell) = \mathbf{r}_{ij}(t+\ell) - \mathbf{r}_{ij}(t)$. The local affine fit is written conceptually as

$$\hat{\mathbf{J}}_i(t) = \arg\min_{\mathbf{J}} \sum_{j\in N_i(t)} \left\| \Delta\mathbf{r}_{ij}(t,\ell) - \mathbf{J}\mathbf{r}_{ij}(t) \right\|^2 . \tag{1}$$

The non-affine residual is the part of neighbor relative motion not explained by this local affine deformation,

$$\mathbf{u}_{ij}^{NA}(t) = \frac{\Delta\mathbf{r}_{ij}(t,\ell) - \hat{\mathbf{J}}_i(t)\mathbf{r}_{ij}(t)}{\Delta t_\ell}. \tag{2}$$

T1 denotes the tangential activity of this focal-neighborhood residual around compact-density transition windows. When the analysis uses unsigned activity, T1 is summarized by the magnitude or squared magnitude of the tangential residual. The essential point is that T1 is not a raw speed, an inferred force, or an individual behavioral rule. It is a local tangential non-affine residual measured after local affine deformation has been removed.

D. Aim and Contribution

This study asks four questions. First, does T1 remain detectable in most observations after local affine subtraction? Second, is that survival stable to nearby choices of local scale and temporal lag? Third, what spatial and temporal form does the surviving residual take? Fourth, are compact state variables, true transition timing, or recent path history sufficient to account for it?

The contribution is deliberately bounded. We do not claim to identify a universal mechanism for midge swarming, nor do we reject existing stochastic or statistical-physics descriptions of these systems. Instead, we define and test a local tangential non-affine residual across 19 laboratory midge-swarm observations. T1 survives local affine subtraction in most observations and is robust within the survivor class, with diffuse tangential activity as the most stable repeated form. At the same time, the tested compact state-variable, event-timing, and recent-history explanations do not account for it consistently across observations. The resulting picture is a reproducible local collective observable with explicit observation-level boundaries, rather than a completed mechanism model.

II. Data and Observables

A. Trajectory Dataset

We analyzed the published three-dimensional trajectories of 19 laboratory *Chironomus riparius* swarms sampled at 100 Hz [8]. Each observation contains time-resolved individual identifiers, positions, velocities, and accelerations. The original experimental setup, imaging system, and reconstruction procedure are described in the dataset paper.

The observations differ in duration, number of tracked individuals, track length, and recording context. In this paper, these differences are treated as part of the empirical domain of the result. They are not removed by pooling, and they are not used as causal explanations unless independently verified. The 19 observations are separate experimental recordings of the same laboratory swarm system rather than 19 independent biological populations. We therefore used observation identity as the replication and grouping unit throughout the cross-observation analyses.

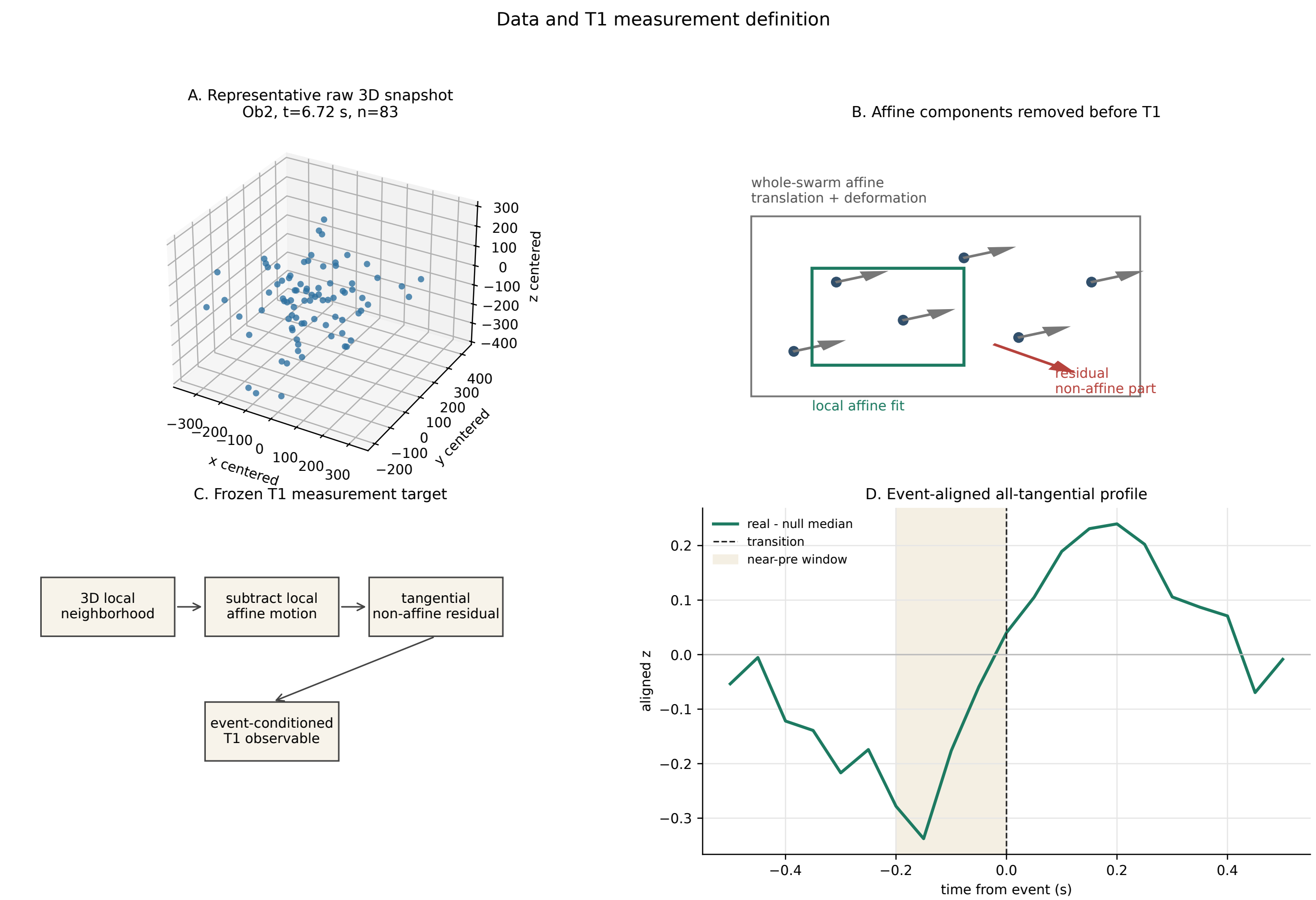

Fig. 1. Data and T1 measurement definition. Panel A shows a representative raw three-dimensional snapshot from one observation. Panels B and C define the affine subtraction and the frozen T1 measurement target used in this study. Panel D shows the event-aligned all-tangential profile used to illustrate the measurement pipeline. The figure defines the observable and does not represent a fitted physical mechanism.

B. Compact-Density State and Transition Events

For each frame, the swarm center and root-mean-square radius were computed as

$$R_{\rm rms}(t) = \sqrt{\frac{1}{n(t)}\sum_i \|\mathbf{x}_i(t) - \bar{\mathbf{x}}(t)\|^2}. \tag{3}$$

We then used the radius-based density proxy

$$\rho_{\rm rms}(t) = \frac{n(t)}{(4/3)\pi R_{\rm rms}(t)^3}. \tag{4}$$

Within each observation, $\rho_{\rm rms}(t)$ was robust-z standardized to define the compact-density coordinate $C(t)$. A one-second centered rolling mean of this standardized coordinate provided the smoothed compact-density trace, and $\dot{C}(t)$ was computed as the time gradient of that smoothed trace. The swarm-size coordinate $R(t)$ was the robust-z standardized $R_{\rm rms}(t)$.

Transition events were inherited from the upstream compact-density coarse-graining. The continuous quantity was $C(t)$ and the low/high labels were the frozen `spectral_set` state labels. A transition was defined as a switch between the two labels. It was retained only when both the preceding and following state runs lasted at least 0.20 s, a persistence screen used to remove single-frame or very short label flicker at the 100 Hz sampling rate. The event time was the first frame of the new run, and events were labeled as low-to-high or high-to-low transitions. The `spectral_set` labels came from an upstream transfer-operator compact-density coarse-graining of robust-standardized `r_rms`, `density_rms`, and `anisotropy`. The selected `eig2` partition separated a more compact high-density state from a less compact low-density state and was constructed upstream of T1, not fitted from T1.

C. Local Non-Affine T1 Observable

T1 is the frozen local tangential non-affine measurement target used in this study. It is not a raw individual velocity, an inferred force, or a behavioral rule. It is computed from focal-neighborhood relative motion after local affine deformation has been removed.

For a focal individual i, let

$$\mathbf{r}_{ij}(t) = \mathbf{x}_j(t) - \mathbf{x}_i(t), \qquad \Delta\mathbf{r}_{ij}(t,\ell) = \mathbf{r}_{ij}(t+\ell) - \mathbf{r}_{ij}(t), \tag{5}$$

where j indexes retained nearest neighbors of the focal individual and ℓ is the finite temporal lag. The local affine deformation was fitted by equal-weight least squares,

$$\hat{\mathbf{J}}_i(t) = \arg\min_{\mathbf{J}} \sum_{j\in N_i(t)} \|\Delta\mathbf{r}_{ij}(t,\ell) - \mathbf{J}\mathbf{r}_{ij}(t)\|^2 . \tag{6}$$

Here, $\hat{\mathbf{J}}_i(t)$ is an affine map of finite-lag relative displacements, not a constitutive deformation tensor or an inferred force-gradient. Because the fit uses focal-centered relative coordinates, local translation is already removed. The finite-lag non-affine residual velocity is

$$\mathbf{u}_{ij}^{NA}(t) = \frac{\Delta\mathbf{r}_{ij}(t,\ell) - \hat{\mathbf{J}}_i(t)\mathbf{r}_{ij}(t)}{\Delta t_\ell}, \tag{7}$$

where Δt_ℓ is the elapsed time corresponding to lag ℓ.

Let $\hat{\boldsymbol{\rho}}_i(t)$ denote the unit vector from the swarm center to focal individual i. Focal samples with non-finite or numerically zero swarm-centered radius were excluded before this projection. The tangential residual is

$$\mathbf{u}_{ij,\tan}^{NA}(t) = \mathbf{u}_{ij}^{NA}(t) - \left[\mathbf{u}_{ij}^{NA}(t)\cdot\hat{\boldsymbol{\rho}}_i(t)\right]\hat{\boldsymbol{\rho}}_i(t). \tag{8}$$

The primary positive analyses used unsigned tangential magnitude or activity. For the later state-matched analyses, the focal-centered activity was frozen as

$$A_{\tan,i}^{NA}(t) = \mathrm{median}_{j\in N_i(t)} \left\|\mathbf{u}_{ij,\tan}^{NA}(t)\right\|^2, \tag{9}$$

and the swarm-level frame activity was

$$A_{\tan}^{NA}(t) = \mathrm{median}_i\, A_{\tan,i}^{NA}(t). \tag{10}$$

We refer to this residual activity family as T1. Signed quantities were used only for secondary event-type and moment-structure tests.

III. Methods

The analysis was organized as a sequence of reduction tests. Each step asked whether a natural explanation was sufficient before interpreting the remaining T1 activity as a reproducible local phenomenon. All manuscript-level claims were evaluated at the observation level rather than by treating all frames or focal samples as independent observations.

A. Local Affine Reduction and Event-Conditioned Controls

For each focal individual and frame, candidate neighbors were first restricted to individuals present at both the reference frame and the finite-lag frame. These candidates were then ranked by Euclidean distance at the reference frame. The focal individual was excluded, the nearest k retained candidates defined the local neighborhood, and focal samples with fewer than four retained neighbors were discarded. The default lag was 0.10 s. The initial survival analysis used two neighborhood sizes, $k = 8$ and $k = 10$, and later focal-activity analyses used the frozen $k = 8$ representation.

The local affine deformation was fitted by equal-weight least squares to finite-lag relative neighbor displacements. Thus, although the raw trajectory files include velocity fields, the frozen local T1 observable was computed from relative displacement residuals divided by the lag duration. This choice keeps the local affine baseline tied to the same finite-lag neighborhood used to define the residual.

Local frame metrics were robust-z standardized within each observation. Slow trends were removed by subtracting a one-second centered rolling mean, and the resulting residual traces were robust-z standardized again before event-conditioned comparisons. Because this detrending uses a centered window, event-aligned time profiles should be interpreted as descriptive association analyses rather than causal or online prediction analyses.

For the survival and spatial analyses, pre/post summaries used $[-0.20, 0)$ s and $[0, 0.20]$ s windows relative to each transition time. Same-observation non-event controls were generated by sampling one control time per real event, requiring each control center to be at least 0.80 s away from true transition times. Forty non-event replicates were used. A T1 survival call at a given neighborhood scale required all three frozen screens: the local event-control excess `local_event_minus_non_event_direction_z` had to exceed 0.03 robust-z units, the empirical non-event tail fraction `p_non_event_direction_ge_event` had to be no greater than 0.35, and the local/global-affine retention ratio `local_to_b3_direction_ratio` had to be at least 0.30. Here, B3 denotes the upstream global-affine residual baseline, so this ratio required the local residual to remain non-negligible relative to that global-affine reference. The 0.35 tail criterion was a frozen screening component rather than a conventional single-observation significance test. It defined a consistent residual-support call from direction, empirical tail rank, and retention relative to the global-affine residual. Statistical calibration of the resulting all-observation both-scale count was performed separately with the full-pipeline pseudo-event omnibus null. The main all-observation count used a two-scale consistency requirement: survival was counted as both-scale support only when the same observation passed this per-scale screen at both original neighborhood sizes, $k = 8$ and $k = 10$.

Three post-freeze checks were then used to harden the manuscript-facing interpretation without changing the frozen T1 definition. First, a full-pipeline pseudo-event omnibus null reran the same survival gate on same-observation pseudo-event structures that avoided true transition windows. The completed high-replicate run used $B = 1000$ null replicates and 40 non-event controls per replicate-observation. For computational tractability, the run was split into deterministic chunks and merged after completion; this changed only execution, not the frozen pseudo-event design or survival gate. Second, a detrending challenge reran the survival and timing summaries under three fixed variants: robust-z standardization followed by centered one-second rolling-mean subtraction and a second robust-z standardization; the same procedure with a trailing, past-only one-second rolling mean; and a no-rolling variant using robust-z standardization only. Third, local affine conditioning quality control sampled frames across all observations and both original neighborhood scales, recording valid-fit fractions, rank sufficiency, condition numbers, and whether the largest T1 samples were concentrated in ill-conditioned fits.

B. Scale, Spatial, and Timing Analyses

Surviving observations were rechecked under nearby choices of local scale and lag. This analysis was used as a robustness check rather than as a search for an optimal parameter set.

Spatial diagnostics decomposed unsigned T1 activity into diffuse, radial-shell, and local-density summaries. Diffuse activity pooled the tangential residual magnitude across sampled focal neighborhoods. Radial-shell diagnostics divided focal individuals into core, middle, and edge thirds by their swarm-centered radius. Local-density diagnostics used the distance to the k-th retained neighbor as a density proxy and compared dense, middle, and sparse thirds. Edge/core and dense/sparse contrasts were treated as candidate spatial descriptions, not as predefined mechanisms.

Timing analyses used an event-aligned profile from -0.50 s to 0.50 s around each transition, sampled every 0.05 s. Four phase bins were used: early-pre $[-0.50, -0.25)$ s, near-pre $[-0.25, 0.00)$ s, near-post $[0.00, 0.25]$ s, and late-post $(0.25, 0.50]$ s. Signed event-type analyses then compared low-to-high and high-to-low transitions within these phase windows. These signed analyses were treated as boundary tests because they asked whether the unsigned T1 activity also had a stable directional or mirror structure.

C. Low-Dimensional State Sufficiency

After local T1 survival was established, we tested whether compact state variables were sufficient to account for first- or second-moment structure in the local tangential non-affine residual. The grouped validation unit was the observation: models were trained on all but one observation and evaluated on the held-out observation. The baseline was a radius-only binned model. The state model added $C(t)$ and $\dot{C}(t)$ to the radius covariate. Performance was measured by incremental

$$R^2_{\mathrm{inc}} = 1 - \frac{\mathrm{MSE}_{\mathrm{state}}}{\mathrm{MSE}_{\mathrm{radius}}}. \tag{11}$$

A within-observation circular shift of C and $\dot{C}$ provided a temporal state-null comparison.

The radius variable depended on the unit of analysis. In the vector-level moment test, the radius covariate was the focal individual's swarm-centered radius. In the state-matched event-locality and history tests described below, the radius coordinate was the swarm-level robust-z $R_{\mathrm{rms}}(t)$.

D. State-Matched Event-Locality

We next asked whether true transition timestamps carried information beyond continuous compact state. For each true event, near-pre T1 activity over $[-0.25, 0.00]$ s was compared with same-observation non-event frames matched in robust standardized $S(t) = (C(t), \dot{C}(t), R(t))$ space. Candidate controls were ranked by Euclidean distance in this standardized state space, and up to five nearest matched controls were used for each event. Candidate controls were required to be at least 0.75 s from true transition times, and the best match distance had to be no greater than 0.75 for the event to be accepted. Controls were not removed after use, so the same non-event frame could in principle match more than one event. A shifted-event null with 80 within-observation replicates was used to test whether true event timings exceeded a temporal reference. This endpoint-inclusive state-matched window follows the 4100 implementation and is distinct from the half-open phase-bin convention used for event-aligned profiles, where the transition frame was assigned to the near-post bin.

E. Recent-History and Observation Heterogeneity

Finally, we tested whether recent state-path direction supplied information not contained in the current state. Frames were paired within the same observation when their current $S(t)$ states were close but their recent paths through the C–R plane were contrasting. For history window h, the recent path vector was

$$\Delta X_h(t) = [C(t) - C(t-h),\ R(t) - R(t-h)]\,, \tag{12}$$

and the contrast angle was the angle between two such vectors. The primary history window was 0.50 s, the current-state distance threshold was 0.50, the history-angle contrast threshold was 90 degrees, and paired frames were required to be at least 1.0 s apart. The observed history-conditioned separation was compared with within-observation shuffled-history controls.

All cross-observation summaries preserved observation identity. Observation heterogeneity was treated as part of the empirical result, while metadata associations were used descriptively only.

IV. Results

The results are organized according to the four questions posed in the Introduction. We first tested whether T1 remained detectable after local affine subtraction and whether this survival decision was stable to nearby scale and lag choices. We then examined the spatial and event-time form of the surviving residual. Finally, we tested whether compact state variables, true transition timing, or recent state-path history accounted for the residual, and used observation-level heterogeneity to define the empirical scope of the claim.

A. T1 Survived Local Affine Subtraction

The first result was that local affine deformation did not fully absorb the T1 observable. Under the two-scale consistency requirement, T1 survived in 14 of the 19 observations. Under the more permissive any-scale criterion, T1 survived in 15 of the 19 observations (Fig. 2). The two-scale number is the main anchor because it shows that the survival decision was not limited to a single original neighborhood-scale setting.

The all-observation support was also checked against a completed full-pipeline pseudo-event calibration. In the observed data, $N_{\mathrm{both}} = 14$; across $B = 1000$ null replicates, the null

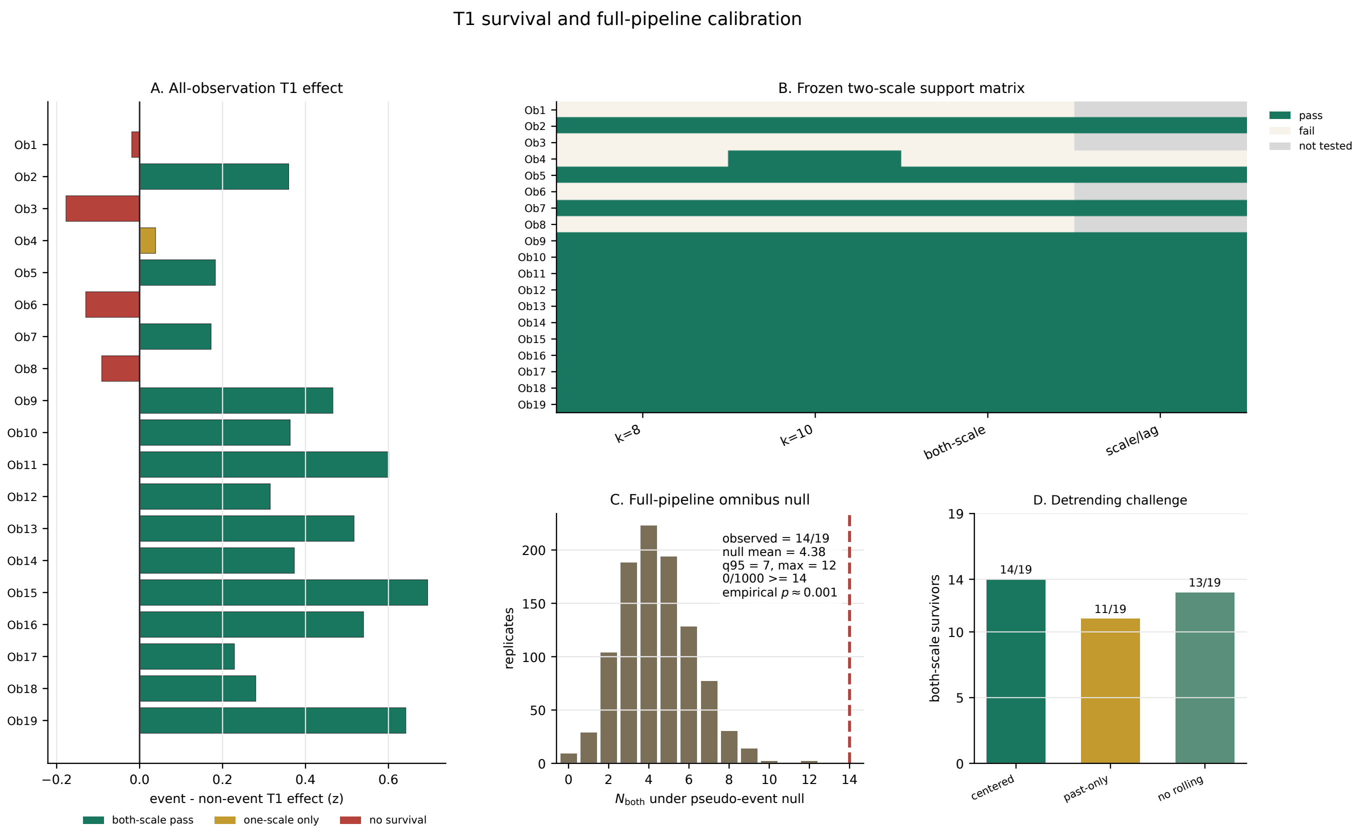

Fig. 2. T1 survival and full-pipeline calibration. The all-observation panel retains both survivor and failure observations, the support matrix shows the frozen two-scale decision structure, the omnibus-null panel shows the distribution of both-scale support across $B = 1000$ pseudo-event replicates, and the detrending panel gives the main preprocessing boundary.

distribution had mean 4.38 and q95 7, and no null replicate reached the observed count of 14 (maximum 12; plus-one empirical $p_{\text{emp}} \approx 0.001$). The any-scale count was supportive but less specific, and is treated as secondary.

The detrending challenge preserved majority-level support but exposed an important boundary. The frozen centered one-second detrending reproduced the main $14/19$ both-scale result. A past-only one-second detrending reduced the stricter both-scale count to $11/19$, while the no-rolling-detrending robust-z variant gave $13/19$. Thus, the T1 survival result was not erased by the challenge. The residual remained detectable at majority level under both alternative preprocessing schemes, while the precise $14/19$ both-scale count was specific to the frozen centered-detrending pipeline.

Local affine conditioning quality control did not indicate that the result was driven by unstable local affine fits. Across all 38 observation-scale combinations, all combinations passed the predefined conditioning screen, all 19 observations passed both k values, the median condition number was 2.37, the largest q95 condition number across combinations was 6.28, and no sampled fit had condition number greater than 100.

The all-observation display is important for interpretation. Several observations were fragile or failed under the predefined criteria. Thus, the result supports common survival after local affine subtraction, not a universal pattern across all observations.

B. Survivor Classifications Were Stable to Nearby Scale and Lag Choices

The second result was that the survivor class was locally stable in parameter space. Of the 15 observations that survived under at least one original scale setting, 14 retained the same qualitative classification under nearby scale and temporal-lag perturbations.

This stability check reduces the possibility that T1 was produced by a narrow or arbitrary parameter choice. Its scope is nevertheless limited: the result supports robustness within the survivor class and within the tested nearby parameter range, rather than robustness for all observations or all possible scales and lags.

C. The Most Repeatable Form Was Diffuse Tangential Activity

After establishing survival and local robustness, we asked what form the surviving residual took across space and event time. The most stable repeated form was diffuse tangential activity: tangential non-affine activity distributed across sampled focal neighborhoods rather than concentrated in a stable edge/core contrast, a fixed near-pre event window, or a universal signed transition direction.

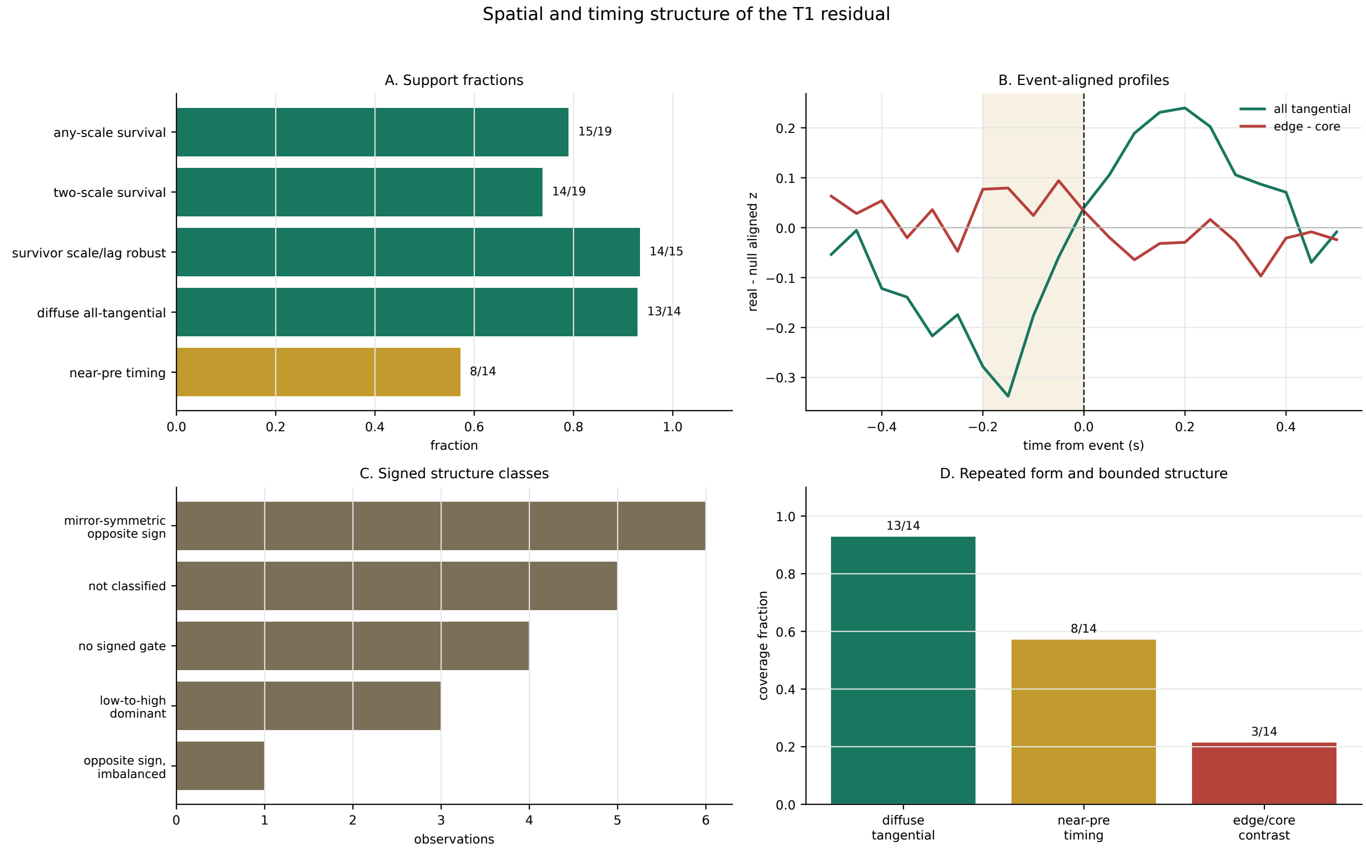

Fig. 3. Spatial and timing structure of the T1 residual. Support fractions and aggregate event-aligned profiles show that diffuse all-tangential activity was the most stable repeated form. Near-pre timing, edge/core contrast, and signed structure were retained as bounded or secondary patterns.

Diffuse all-tangential activity was supported in 13 of the 14 tested observations, whereas near-pre timing was supported in 8 of the 14 tested observations (Fig. 3). Edge/core contrast appeared in a subset of observations, but it was weaker as a phase-specific interpretation. Signed event-type structure also remained heterogeneous and did not define a stable low-to-high, high-to-low, or mirror-symmetric rule. The supported phenotype is therefore distributed local tangential non-affine activity, not a sharp event precursor, stable edge trigger, or global signed law.

The near-pre enrichment should be read with this boundary in mind. It was a descriptive pattern under the original event-control comparison and did not survive the later state-matched event-locality challenge described below. The detrending-specific audit of the same near-pre window retained majority-level support under centered, past-only, and no-rolling variants, but that audit used a looser phase-tail gate and a different pseudo-event sampler. We therefore treat those counts as sensitivity evidence rather than as replacements for the original $8/14$ phase-localization count. Thus, the near-pre profile remained a moderate timing description, not evidence for a causal online precursor.

D. Compact State and Event Timing Did Not Explain the Residual

The compact-state reduction did not support a stable moment closure based on $S(t) = (C(t), \dot{C}(t), R(t))$. The median incremental R^2 was approximately -6.33×10^{-4} for the first-moment target and -6.21×10^{-4} for the second-moment target, with low positive-observation fractions. This result does not rule out all stochastic or state-dependent descriptions. It shows that the tested compact-state representation did not provide a stable first- or second-moment explanation of T1.

The state-matched event-locality test produced a related boundary. True transition timestamps did not show robust near-pre excess beyond same-observation non-event frames matched in $S(t)$. The median event-minus-control near-pre effect was approximately -0.033, and the positive-observation fraction was about 0.42 (Fig. 4). This does not mean that transitions have no special dynamics; it means that the tested state-matched near-pre aggregate did not provide a stable event-specific explanation.

E. Recent History and Observation Heterogeneity Defined the Boundary

Recent state-path history produced an intermediate result. Fourteen of the 19 observations exceeded the shuffled-history

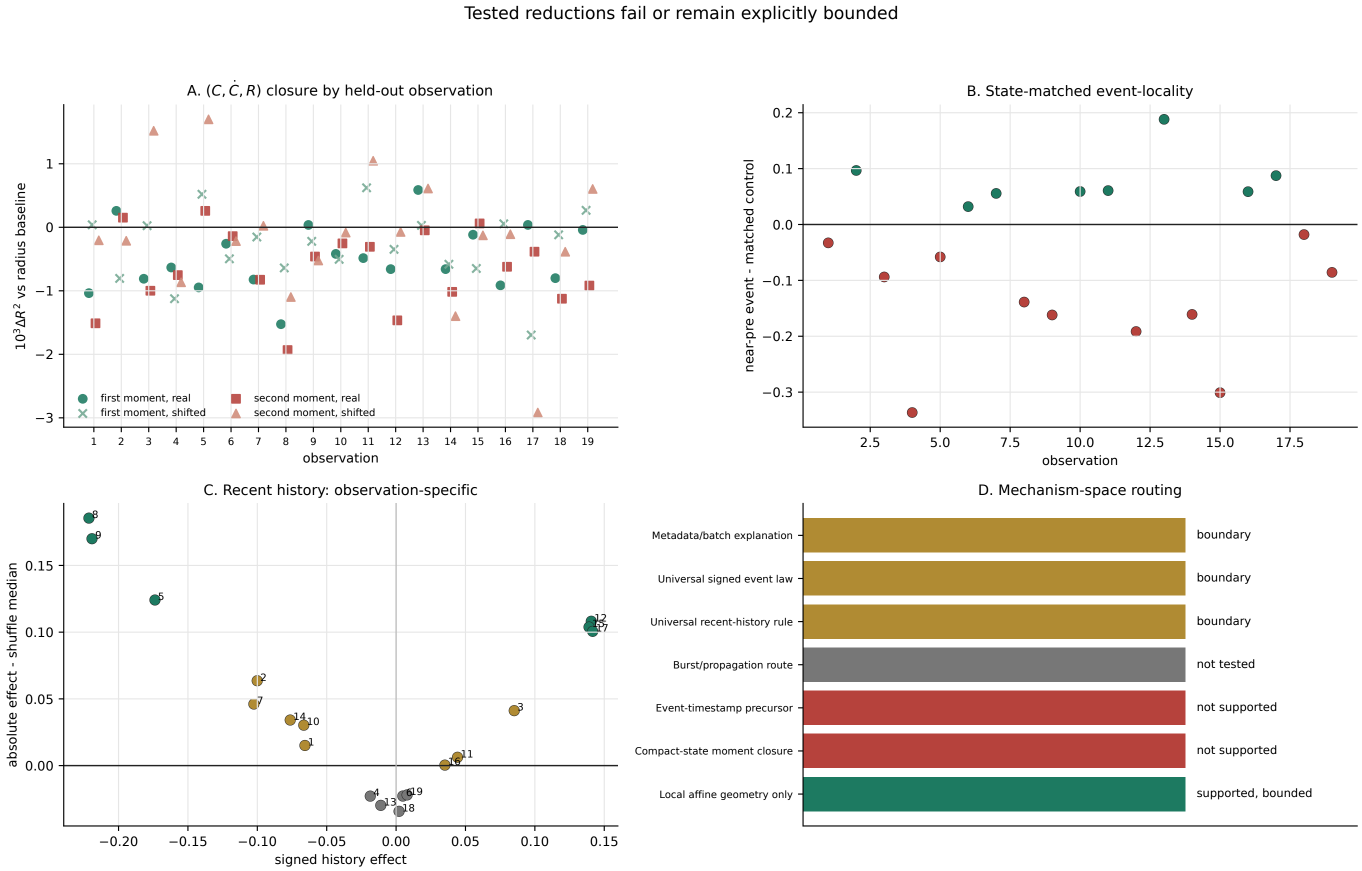

Fig. 4. Reduction boundaries. The tested $S(t) = (C(t), \dot{C}(t), R(t))$ first/second moment closure and the state-matched event-local near-pre test did not provide stable reductions of T1. Recent-history separation was visible in some observations but did not become a universal sign/order rule. Propagation remained outside the current confirmatory route.

median, but only 6 of the 19 exceeded the q95 null threshold, and the sign and ordering of the effect were not stable across observations. Thus, recent history was informative in an observation-specific way, but it did not define a universal memory rule.

The observation-level synthesis showed that heterogeneity was reproducible and observation-specific rather than erased by pooling. Thirteen observations were classified as robust survivors with diffuse support, two as fragile boundaries, two as stable non-survivors, one as a one-scale survivor, and one as a robust survivor without diffuse support (Fig. 5). This heterogeneity constrains the manuscript claim: T1 is common in this dataset, but the support, form, and explanatory status of the residual vary across observations.

V. Discussion and Conclusion

A. *Why the Residual Is Informative*

The interest of T1 is not that it explains how midge swarms are generated or that it defines a universal collective-motion law. Its value is that it makes a specific layer of motion measurable. In biological groups, raw velocity differences across space are expected because groups turn, expand, contract, and rearrange locally. The present result is more specific: a local tangential residual remained measurable after ordinary local affine deformation had been admitted into the baseline and subtracted.

This places T1 between whole-swarm state variables and individual behavioral rules. It is not a raw trajectory feature, and it is not a complete mechanism. It is a bounded local residual layer whose survival, form, and explanatory limits can be tested observation by observation. T1 should therefore not be interpreted as a force, a plastic rearrangement variable in the materials-science sense, or a direct interaction metric. The non-affine construction is used here only as a kinematic diagnostic of local relative motion not represented by the fitted affine map.

The positive and negative results therefore work together. Survival after local affine subtraction establishes a phenomenon worth explaining. The failure of the tested compact-state, event-timing, and recent-history reductions shows that several natural simple explanations were insufficient under their current definitions. The study thus identifies a measurable empirical target and the boundaries of several candidate reductions.

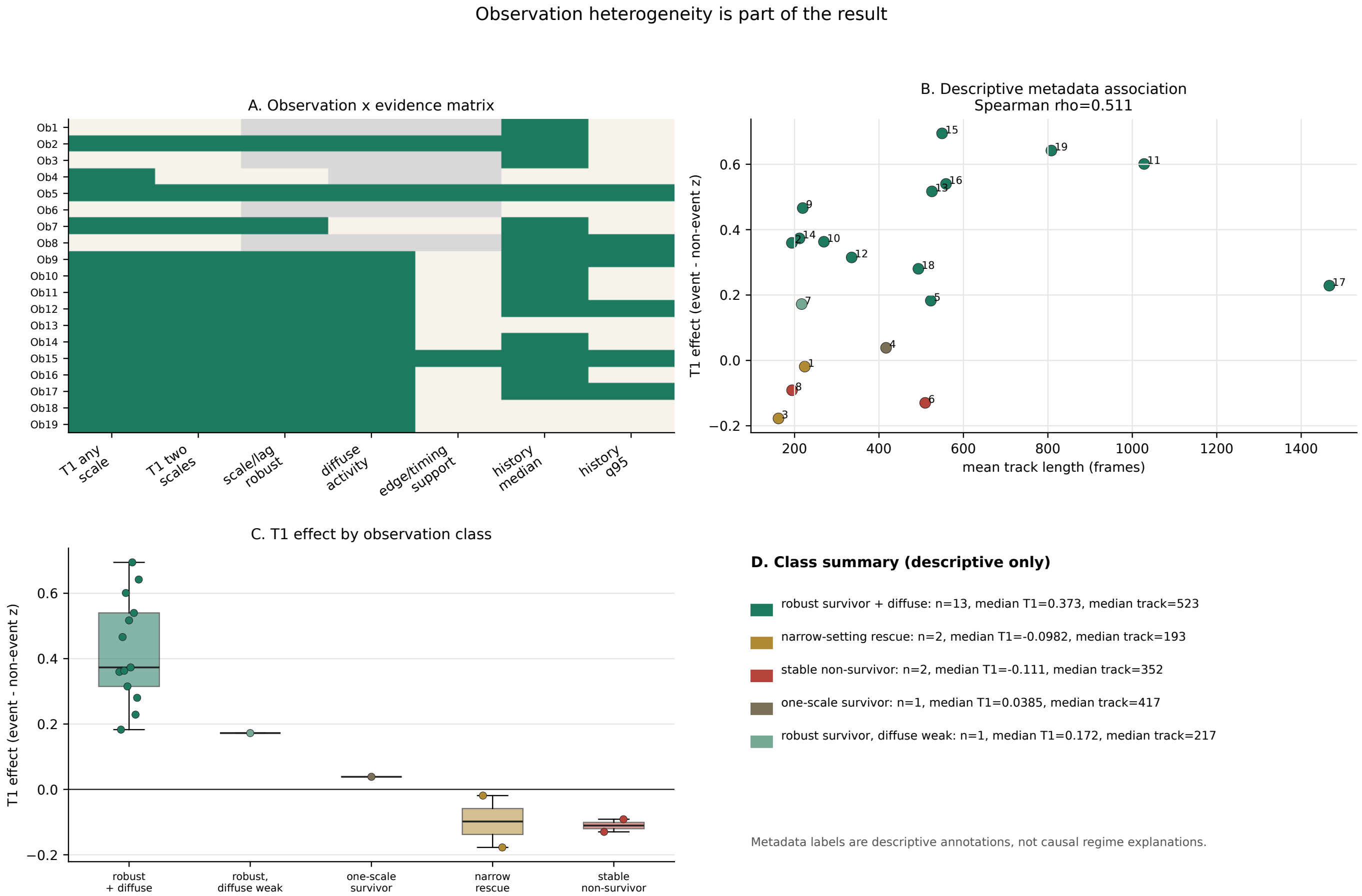

Fig. 5. Observation heterogeneity. Route-level evidence and observation classes show robust survivors, fragile boundaries, and stable failures across the 19 observations. The metadata association panel is descriptive and leave-one-observation-out audited, but it is not a causal recording-condition explanation.

B. Relation to Midge-Swarm and Collective-Motion Studies

The interpretation is consistent with previous midge-swarm studies that treat these systems as weakly ordered nonequilibrium collectives with spatial structure, stochastic components, and effective interactions [1], [3]–[7]. Macroscopic and statistical-physics descriptions make compact state variables natural explanatory candidates, while Langevin-type studies caution against a purely deterministic trajectory interpretation. The present analysis adds a local residual diagnostic: for T1, the tested compact-state representation did not provide a stable reduction.

The non-uniform spatial sampling result is a useful point of comparison [1]. That work made a hidden layer of swarm organization measurable by showing that individuals do not sample the swarm volume uniformly. The present analysis makes a different hidden layer measurable by asking what local motion remains after ordinary local affine geometry has been subtracted. In both cases, the contribution is to separate structure that can be obscured by raw trajectories or whole-swarm averages.

The same diagnostic problem may arise in other collective-motion systems, such as fish schools, bird flocks, cell collectives, or artificial groups. Raw velocity differences in such systems can reflect whole-group motion, local deformation, or interaction-driven residuals. Local affine subtraction may therefore provide a conservative baseline for future comparative studies, although the present data do not show that the same T1 observable appears outside midge swarms. More generally, the framework provides a conservative way to distinguish ordinary neighborhood deformation from residual collective motion in weakly ordered groups. Because mating swarms remain cohesive without flock-like polarization, identifying residual local organization after geometric subtraction helps specify where biologically relevant coordination may reside when conventional global order parameters are weak.

C. Limitations and Future Work

Several limitations define the interpretation of this study. First, the dataset contains 19 laboratory observations, which support common-survival and observation-boundary claims but not a universal mechanism claim. These observations are separate experimental recordings rather than independent biological populations. Second, T1 is a constructed residual depending on neighborhood selection, finite lag, local affine fitting, and tangential projection. It is not a directly observed force, intention, or individual decision variable. Third, the

TABLE I

EVIDENCE-TO-INFERENCE SUMMARY. EACH SUPPORTED INFERENCE IS BOUNDED BY THE TESTED DEFINITION AND SHOULD NOT BE UPGRADED BEYOND THE INTERPRETIVE BOUNDARY LISTED IN THE FINAL COLUMN.

Test	Evidence	Supported inference	Interpretive boundary
Local non-affine survival	14/19 both-scale and 15/19 any-scale survival; full-pipeline pseudo-event calibration $p_{\mathrm{emp}} \approx 0.001$ at $B = 1000$	T1 is common after local affine subtraction	Does not establish universality, causality, or a completed mechanism
Detrending challenge	Centered 14/19, past-only 11/19, no rolling detrending 13/19	T1 survival is not erased by the challenge	The exact 14/19 count is not preprocessing-invariant
Affine-fit conditioning QC	38/38 observation-scale combinations passed; median condition number 2.37	Local affine subtraction is numerically defensible in the sampled QC	Does not rule out all preprocessing artifacts
Scale/lag robustness	14/15 survivor observations robust	Survivor-class robustness is high	Does not establish robustness for all 19 observations
Diffuse tangential form	13/14 diffuse all-tangential support	Diffuse tangential activity is the strongest repeated form	Does not identify a universal edge trigger or signed force
State moment closure	Median incremental R^2 near zero	The tested $S(t)$ closure is not supported	Does not rule out richer stochastic dynamics
Event-locality	Median near-pre event-minus-control effect about -0.033	The tested state-matched near-pre route is not supported	Does not rule out all transition-specific dynamics
Recent history	14/19 above median shuffle, 6/19 above q95	History effects are observation-specific	Does not establish a universal memory rule
Propagation	Confirmatory route not entered	Propagation remains open	Propagation is untested, not rejected

analysis is observational and cannot establish individual-level causality.

The negative tests are also definition-bound. Failure of the tested $(C, \dot{C}, R)$ closure does not rule out higher-dimensional, delayed, network-based, or observation-specific stochastic descriptions. Failure of the state-matched near-pre test does not imply that transitions have no special dynamics. Propagation was not tested in a confirmatory route and therefore remains outside the current supported claim.

The submission-hardening checks sharpened these limitations rather than removing them. The completed full-pipeline pseudo-event calibration supported the survival result: the observed both-scale count of 14 exceeded the null maximum of 12 across $B = 1000$ replicates; equivalently, no null replicate reached 14, giving a plus-one empirical $p_{\mathrm{emp}} \approx 0.001$. This strengthens the observation-level null calibration, but it does not remove the other boundaries. The detrending challenge showed that the signal was not erased, but the exact both-scale count depended on the preprocessing choice: centered detrending gave $14/19$, past-only detrending gave $11/19$, and no rolling detrending gave $13/19$. The local affine conditioning quality control was reassuring, with all 38 observation-scale combinations passing and no sampled condition number above 100, but this is a numerical QC result rather than a biological mechanism result.

Future work should proceed along two broad directions. First, comparative applications should test whether local affine subtraction reveals similar residual layers in other weakly ordered collectives. Second, perturbation experiments or richer measurements of environmental and interaction variables are needed to determine what biological processes generate T1. Observation stratification within this dataset can guide both directions by identifying robust survivors, fragile cases, and stable failures.

D. Conclusion

This study identified a local tangential non-affine residual that remained measurable after local affine deformation was removed from laboratory midge-swarm trajectories. The result is not a complete mechanism: the residual was common but not universal, was rare under a completed full-pipeline pseudo-event calibration, passed numerical conditioning checks, remained partly sensitive to detrending choice, and was not consistently accounted for by the tested compact-state, event-timing, and recent-history reductions.

Its value is diagnostic. T1 marks a level of collective motion between raw individual trajectories and low-dimensional swarm state, and it provides a bounded target for future comparative and mechanistic work.

DATA AND CODE AVAILABILITY

The raw three-dimensional midge-swarm trajectories analysed in this study are external data from the published dataset of Sinhuber et al. [8] and are not redistributed here. Analysis code, frozen configurations, derived summary outputs, final manuscript artifacts, and the completed high-replicate pseudo-event calibration are available at https://github.com/Saru1228/midge-swarm-nonaffine-residuals, release tag `v4158-review010-polish`. The repository includes a lightweight verification script that checks the frozen manuscript package and the $B = 1000$ omnibus-null result without requiring access to the raw trajectories.

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